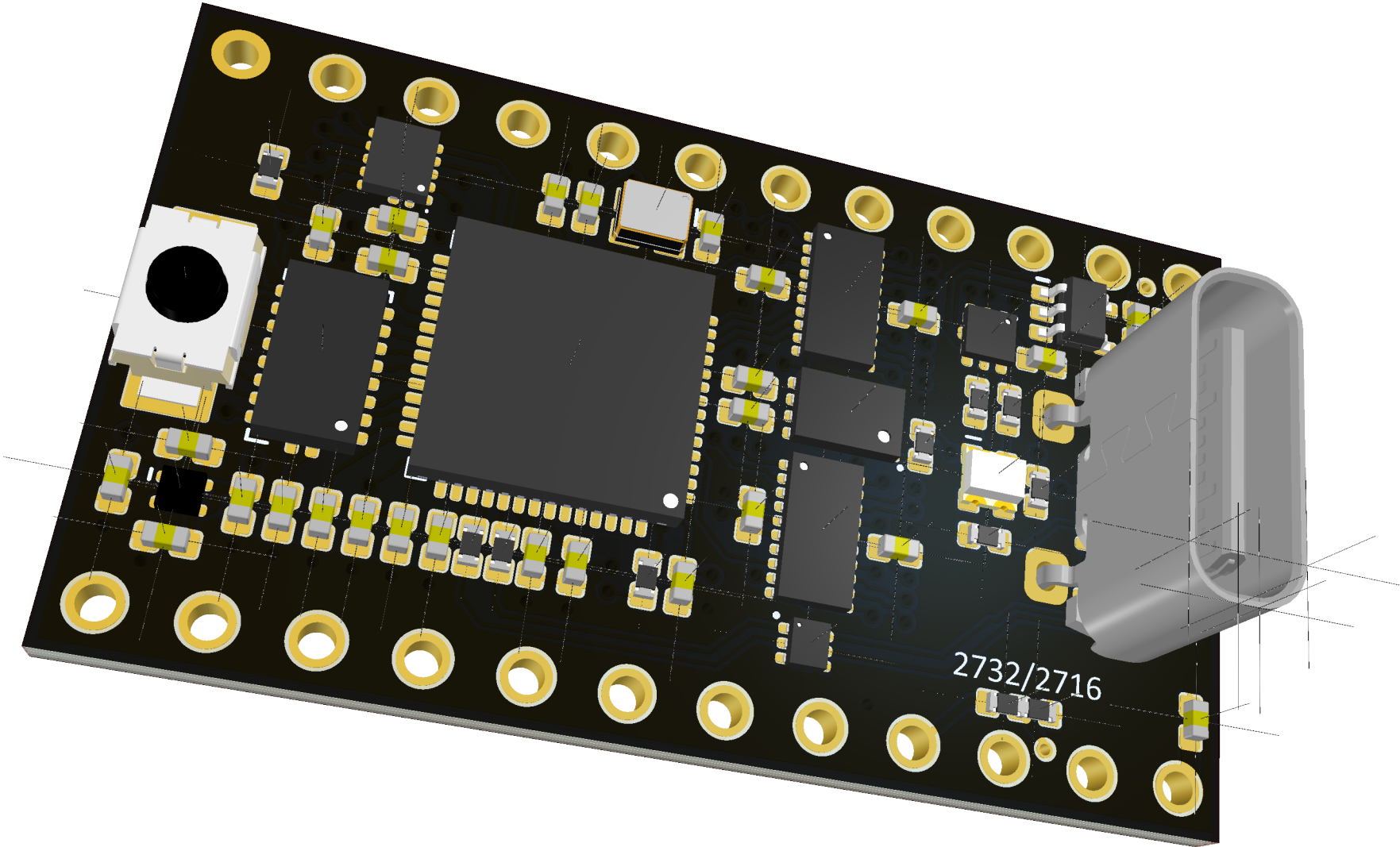


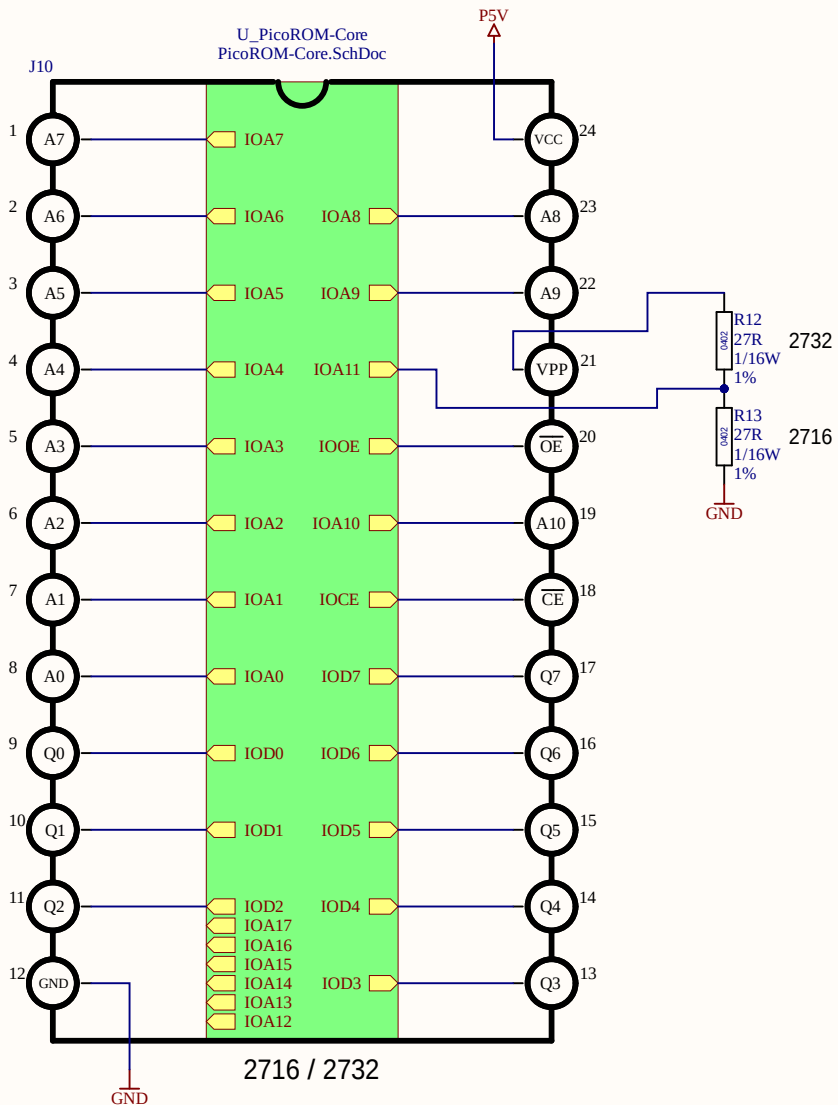
PicoRom-2732.PrjPcb

Realistic View



Document Creation Date: 9/27/2025

Design : Vincent Himpe

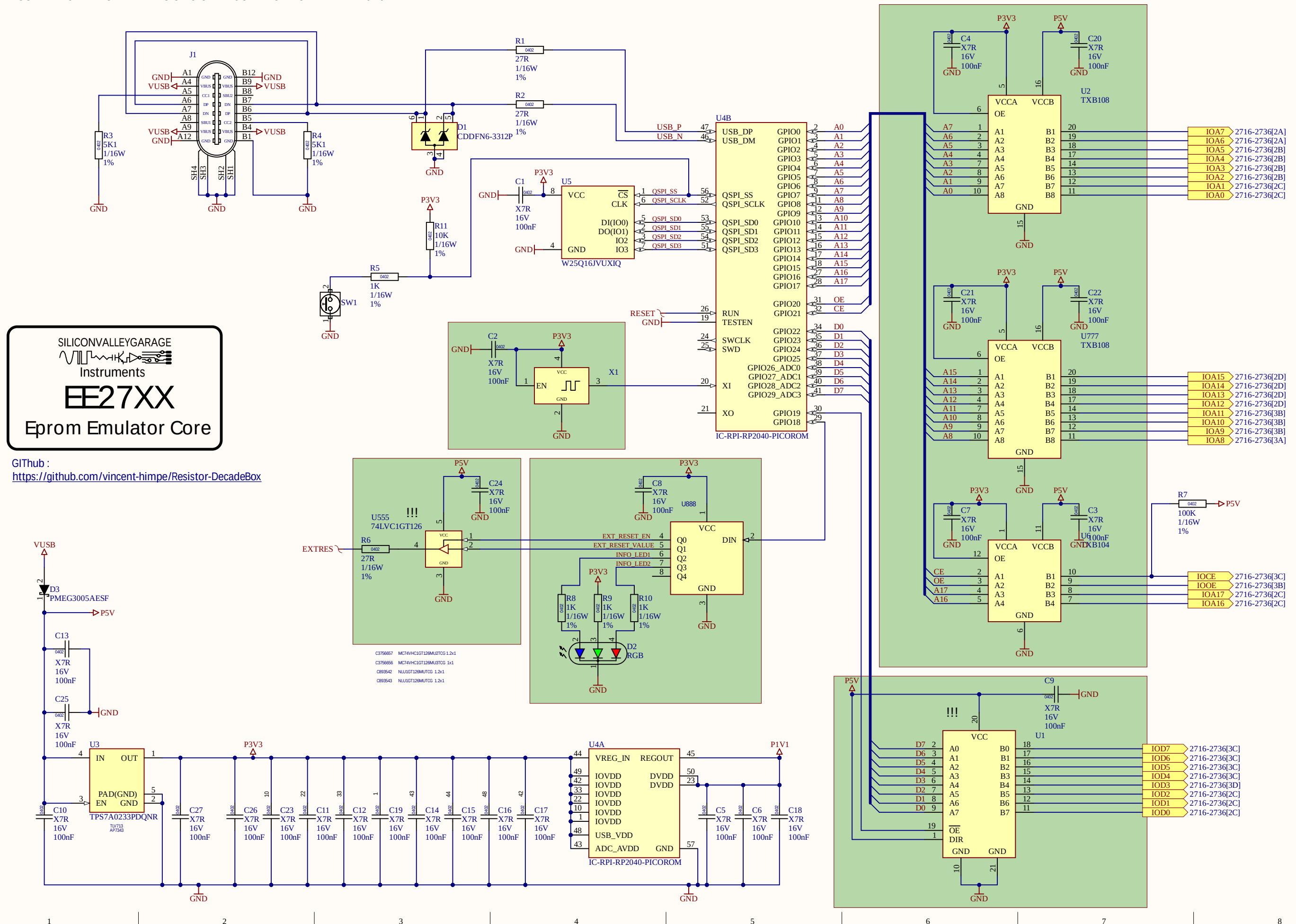


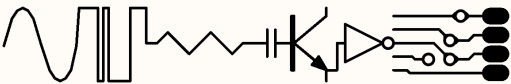
SILICONVALLEYGARAGE
Instruments

EE2732
Eeprom Emulator

GitHub :
<https://github.com/vincent-himpe/Resistor-DecadeBox>

!!! DO NOT DEVIATE FROM THE PARTS !!!
THESE COMPONENTS, ALTHOUGH CMOS, HAVE TTL LEVEL INPUTS (2~2.4V Hi!)
SO THEY CAN WORK WITH 3.3V SIGNAL COMING FROM THE RP2040



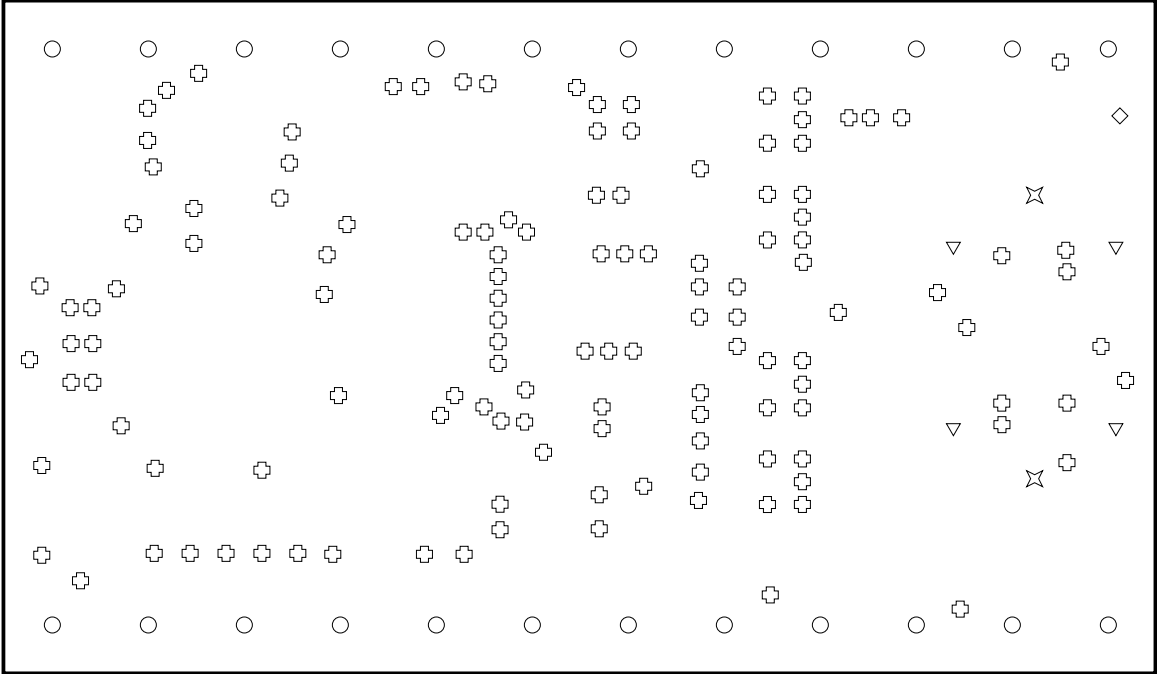
1	2	3	4	5	6
GENERAL					
A	GENERAL 1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA 2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED. 3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER 4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2 5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED 6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
	MATERIALS 7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT. 8. MATERIAL SHALL HAVE A FLAMABILITY RATING OF UL 94V-0 OR BETTER 9. SURFACE FINISH : HASL 10. SOLDER MASK COLOR : BLACK 11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm 12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL 13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG 14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
D	QA, ELECTRICAL TEST AND MARKINGS 15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE 				Project PicoRom-2732.PrjPcb Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6

Drill Table

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via / Pad	Pad Shape	Description	Hole Tolerance	Via Type	Via Feature
⊕	130	0.300mm(11.811mil)	Plated	Round	Top Layer - Bottom Layer	Via				(Mixed)	(Mixed)
✕	2	0.520mm(20.472mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
▽	4	0.600mm(23.622mil)	Plated	Slot	Top Layer - Bottom Layer	Pad	Rounded Rectangular				
◇	1	0.800mm(31.496mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
○	24	0.900mm(35.433mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	161 Total										

DRILL DRAWING

Drill Drawing View (Scale 5:1)



COMPOSITE VIEW FRONT

A

A

B

B

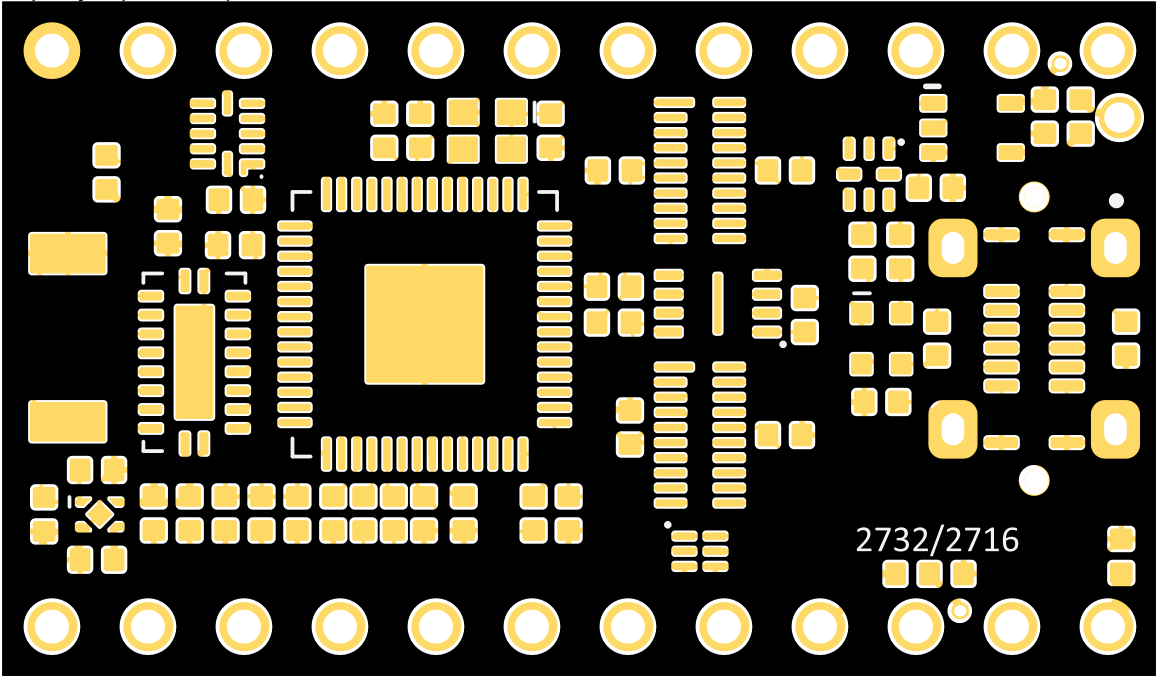
C

C

D

D

Top Layer (Scale 5:1)



SILICONVALLEYGARAGE

Project PicoRom-2732.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

COMPOSITE VIEW BACK

A

A

B

B

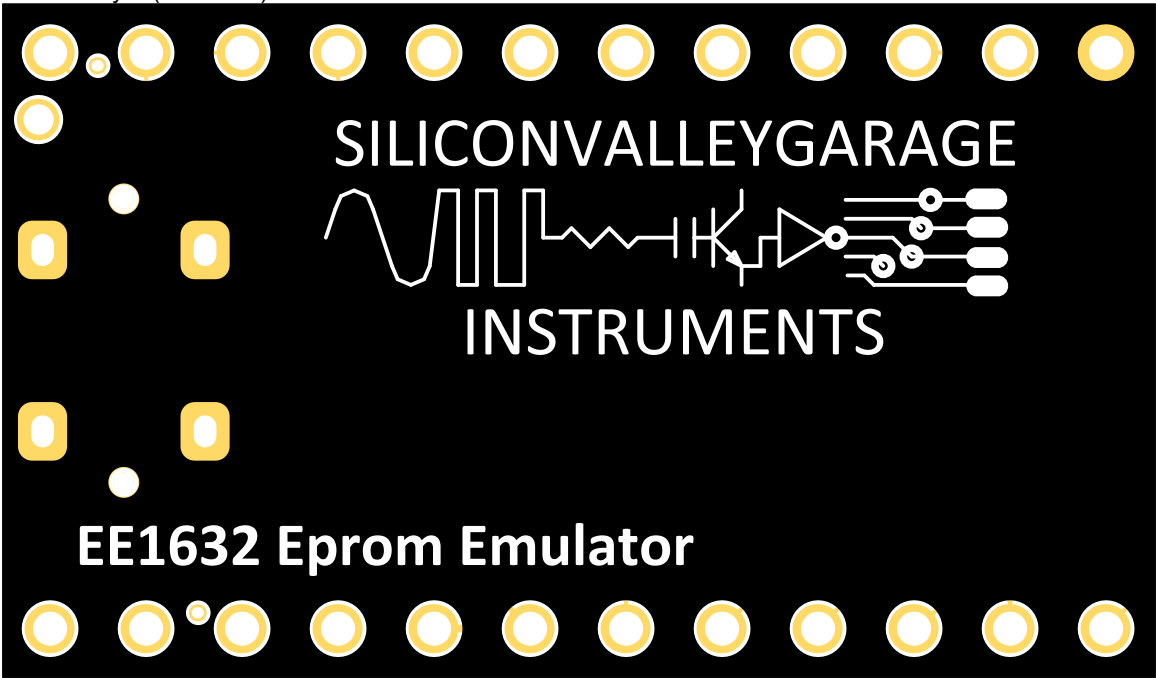
C

C

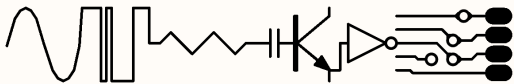
D

D

Bottom Layer (Scale 5:1)



SILICONVALLEYGARAGE

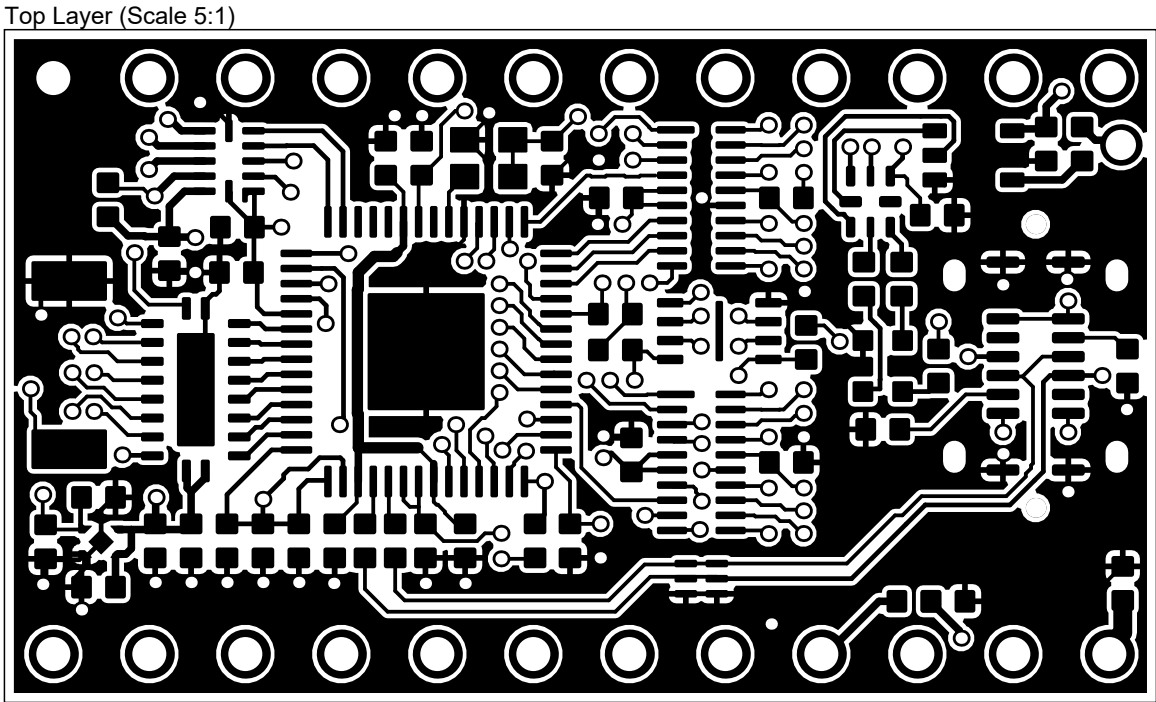


Project PicoRom-2732.PrjPcb

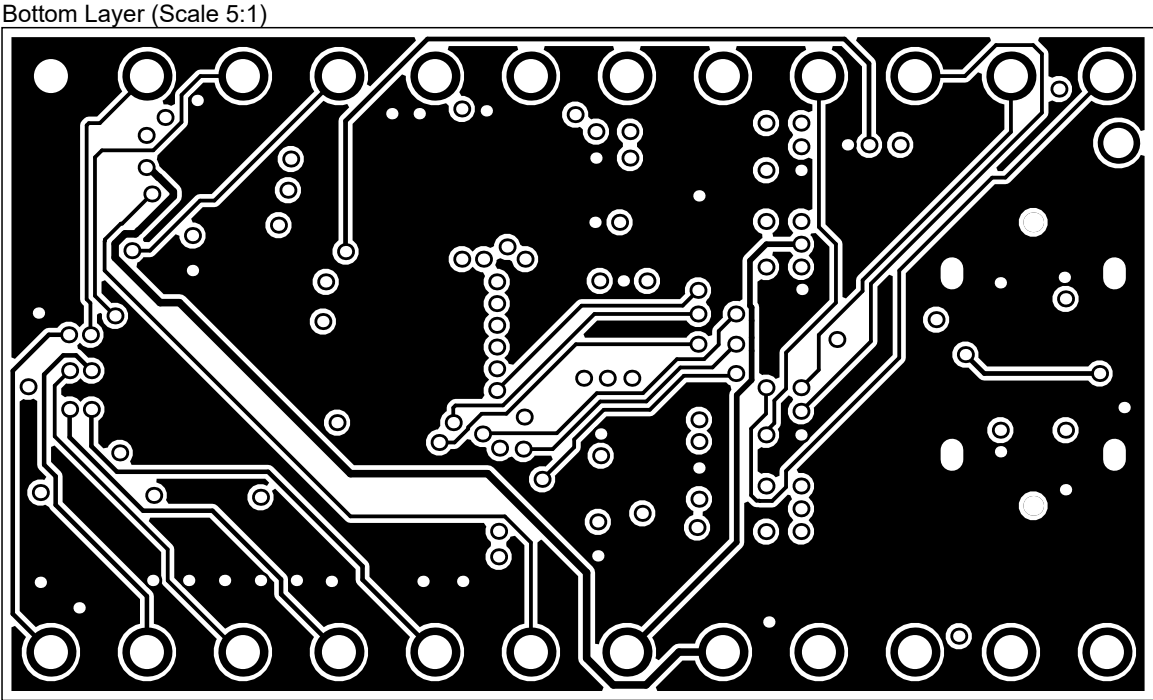
Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : TOP LAYER

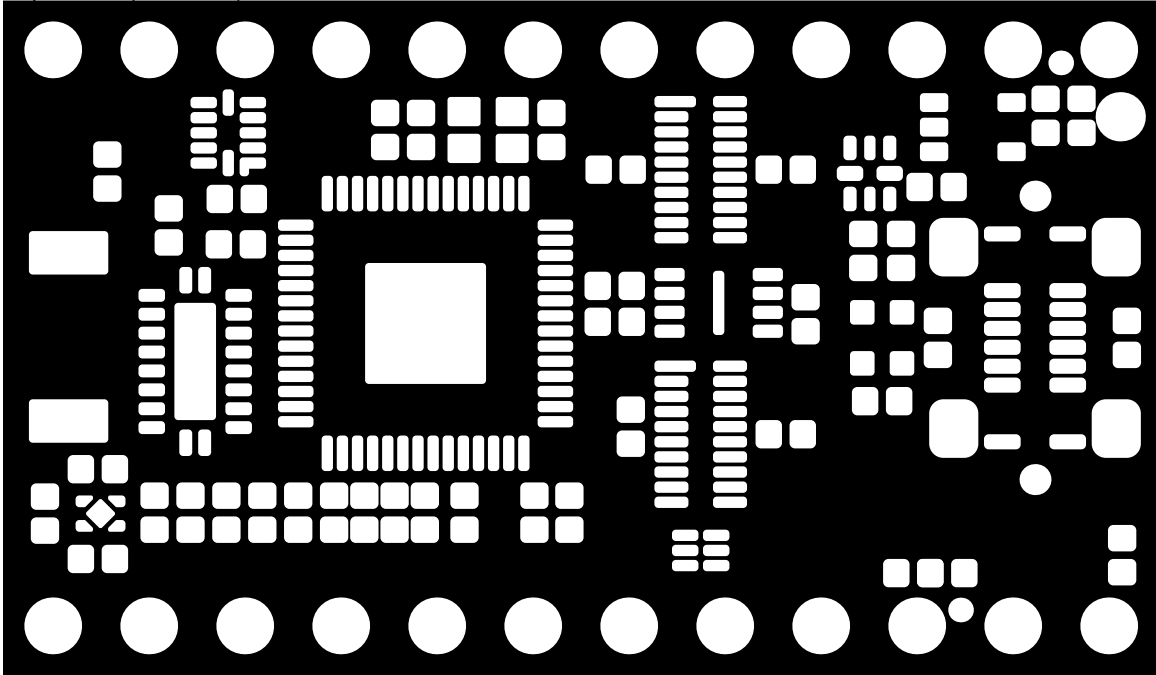


LAYER VIEW : BOTTOM LAYER



LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 5:1)



SILICONVALLEYGARAGE

Project PicoRom-2732.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : BOTTOM SOLDER MASK

A

A

B

B

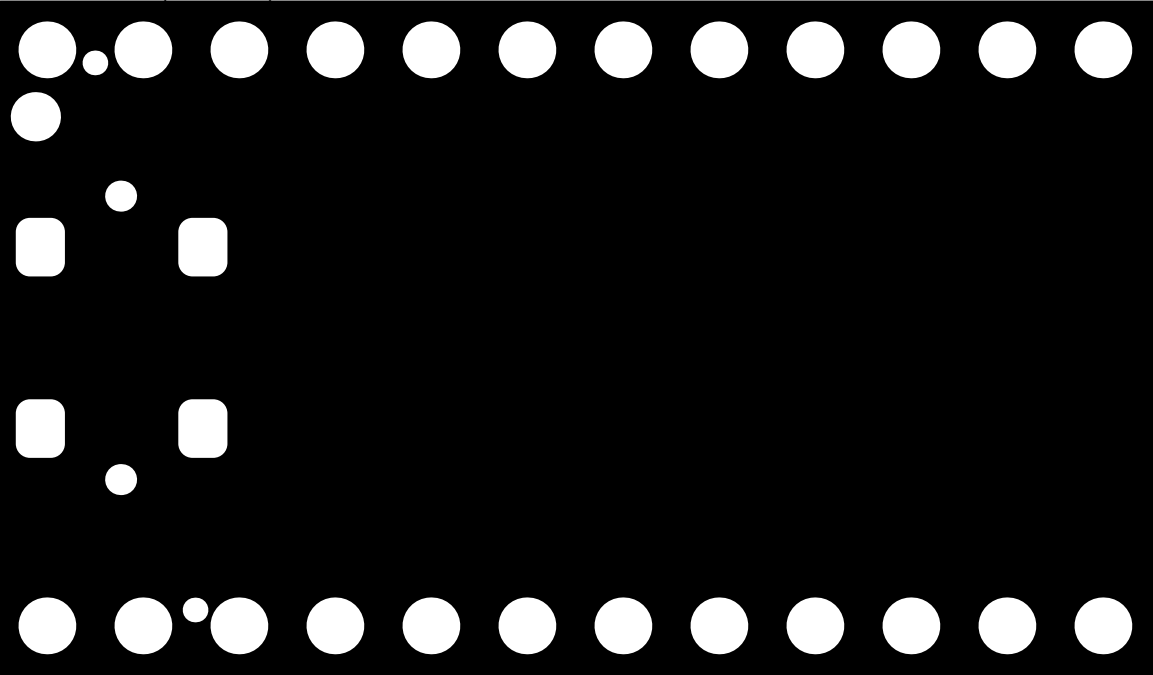
C

C

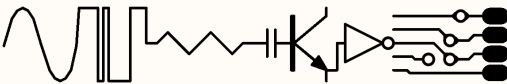
D

D

Bottom Solder (Scale 5:1)



SILICONVALLEYGARAGE

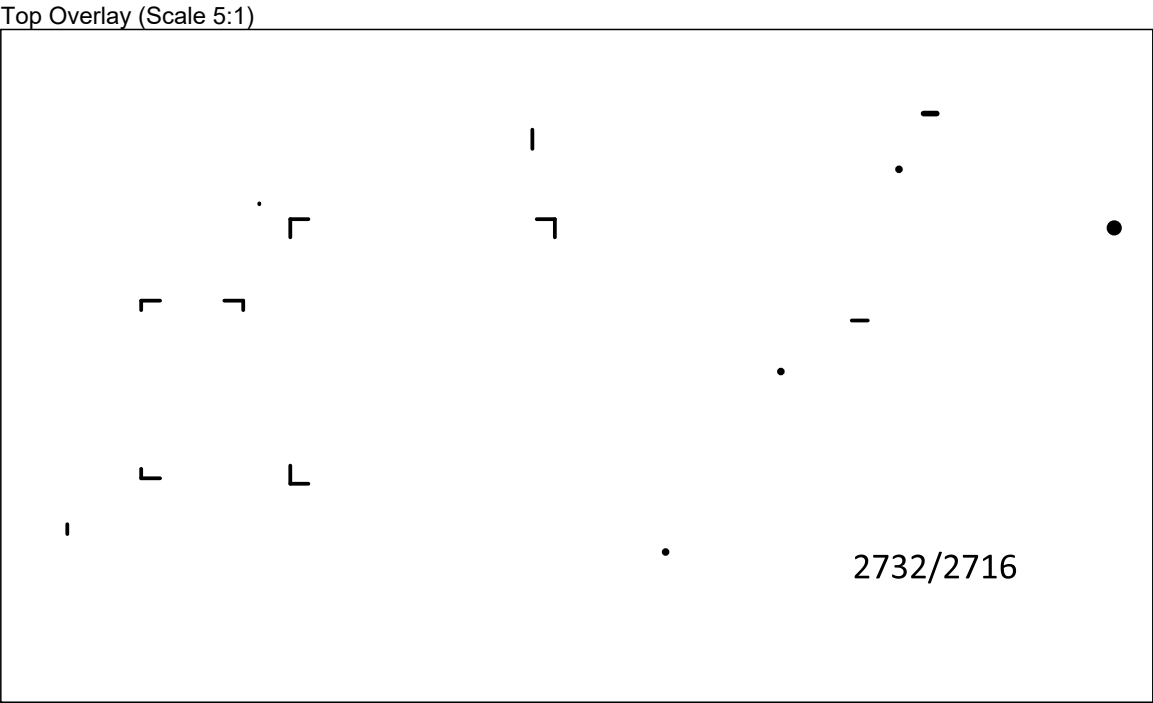


Project PicoRom-2732.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

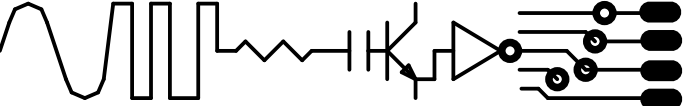
LAYER VIEW : TOP SILKSCREEN (LEGEND)



LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

Bottom Overlay (Scale 5:1)

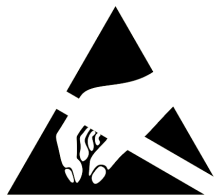
SILICONVALLEYGARAGE



INSTRUMENTS

EE1632 Eprom Emulator

GENERAL



Unless otherwise specified the following rules apply:

- 1. DO NOT DEVIATE FROM ARTWORK OR BOM WITHOUT PRIOR AUTHORIZATION.
- 2. ASSEMBLE AND INSPECT PER IPC-610 CLASS 2

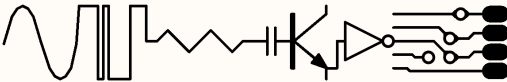
Bill of Materials and Material Handling

- 3. THE BOM CONTAINED IN THIS DOCUMENT IS AS-BUILT. NON-INSTALLED PARTS HAVE BEEN REMOVED. ADDITIONAL BOM FORMATS ARE AVAILABLE IN THE PROJECT FILES
- 4. ANY PART SUBSTITUTIONS MUST BE APPROVED IN WRITING BEFORE ASSEMBLY
- 5. ALL MATERIALS MUST BE PROCURED FROM MANUFACTURER AUTHORIZED DISTRIBUTORS OR THE ORIGINAL MANUFACTURER
- 6. ALL COMPONENTS AND BOARDS TO BE HANDLED AND STORED ACCORDING TO IPC GUIDELINES
- 7. ESD CONTROL PER IPC RULES

Soldering

- 8. SOLDERING TO BE DONE USING SN37PB63 ALLOY USING ALLOY MANUFACTURER RECOMMENDED NO-CLEAN FLUX
- 9. BGA COMPONENTS WITH LEAD-FREE CONNECTIONS NEED TO BE REBALLED WITH SN63PB37. MIXING OF ALLOYS IS NOT PERMITTED.
- 10. SOLDERING PREFERRABLY TO BE DONE USING NITROGEN ATMOSPHERE
- 11. SURPLUS COMPONENTS TO VACUUM SEALED WITH DESSICANT IN ANTISTATIC BAGS
- 12. INCOMING MATERIAL (BOARDS AND COMPONENTS) NEEDS TO BE INSPECTED FOR HUMIDITY AND BAKED IF NEEDED PRIOR TO USE.
- 13. MANUAL REWORK / TOUCHUP TO BE DONE USING SAME ALLOY AND APPROPRIATE FLUX. FLUX MUST BE REMOVED.

SILICONVALLEYGARAGE



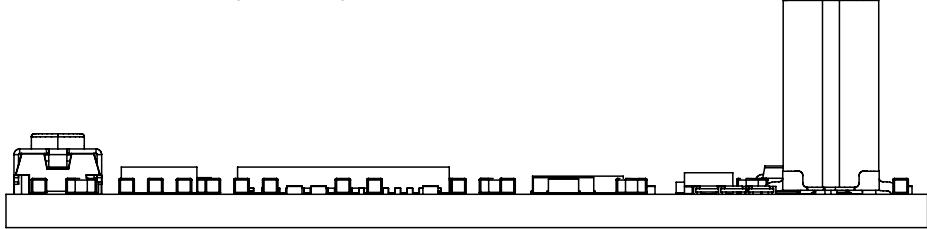
Project PicoRom-2732.PrjPcb

Version: | Variant [No Variations]

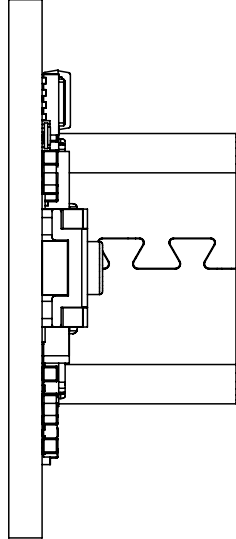
ASSEMBLY DRAWING

2D VIEW

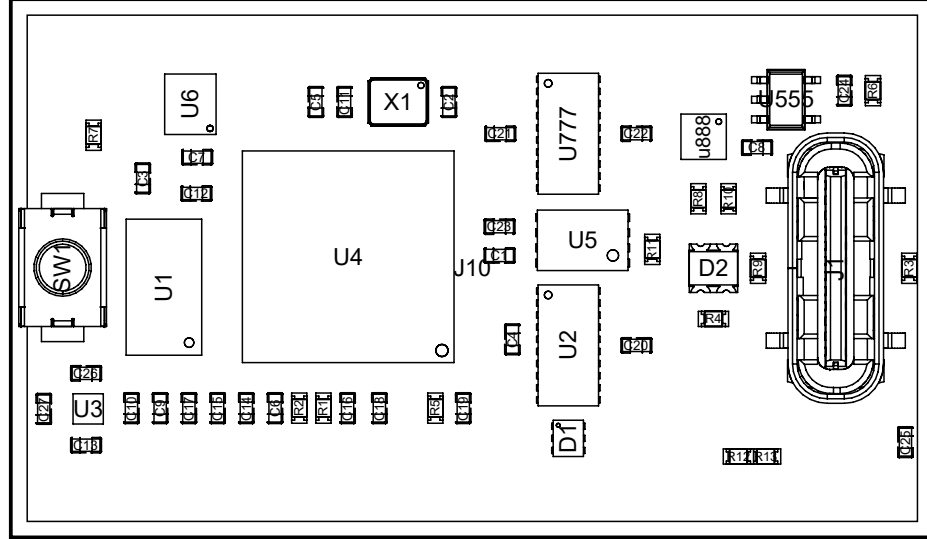
View from Front side (Scale 4:1)



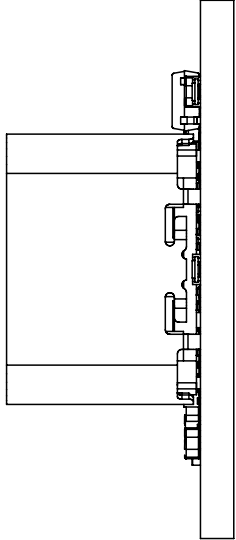
View from Left side (Scale 4:1)



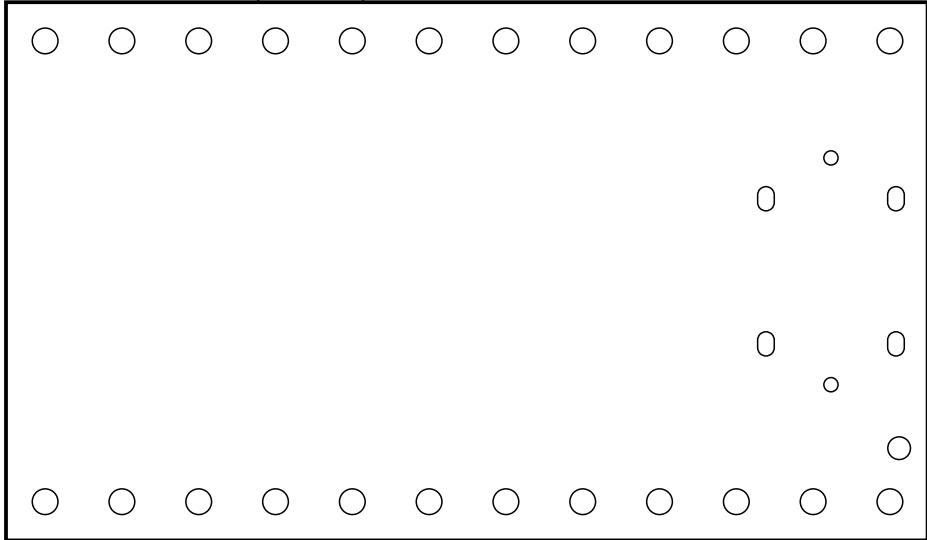
View from Top side (Scale 4:1)



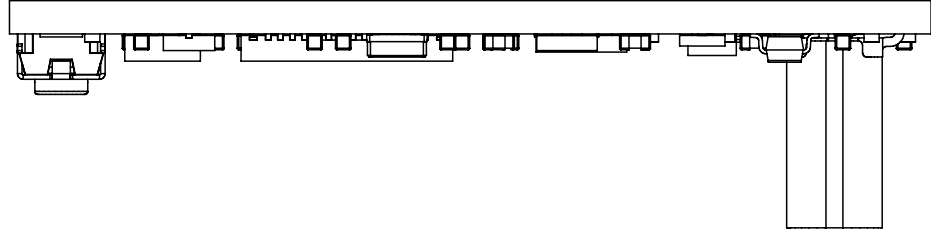
View from Right side (Scale 4:1)



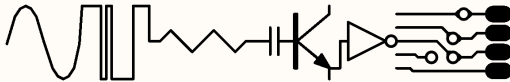
View from Bottom side (Scale 4:1)



View from Back side (Scale 4:1)



SILICONVALLEYGARAGE



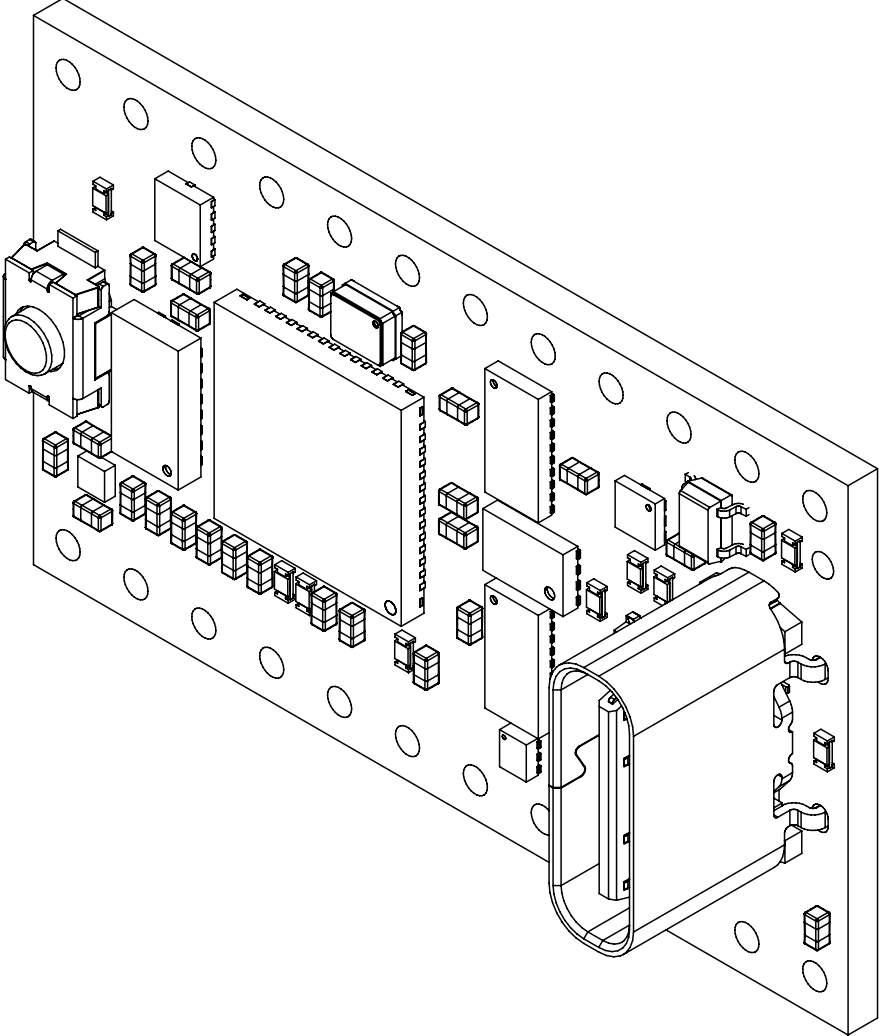
Project PicoRom-2732.PrjPcb

Version: | Variant [No Variations]

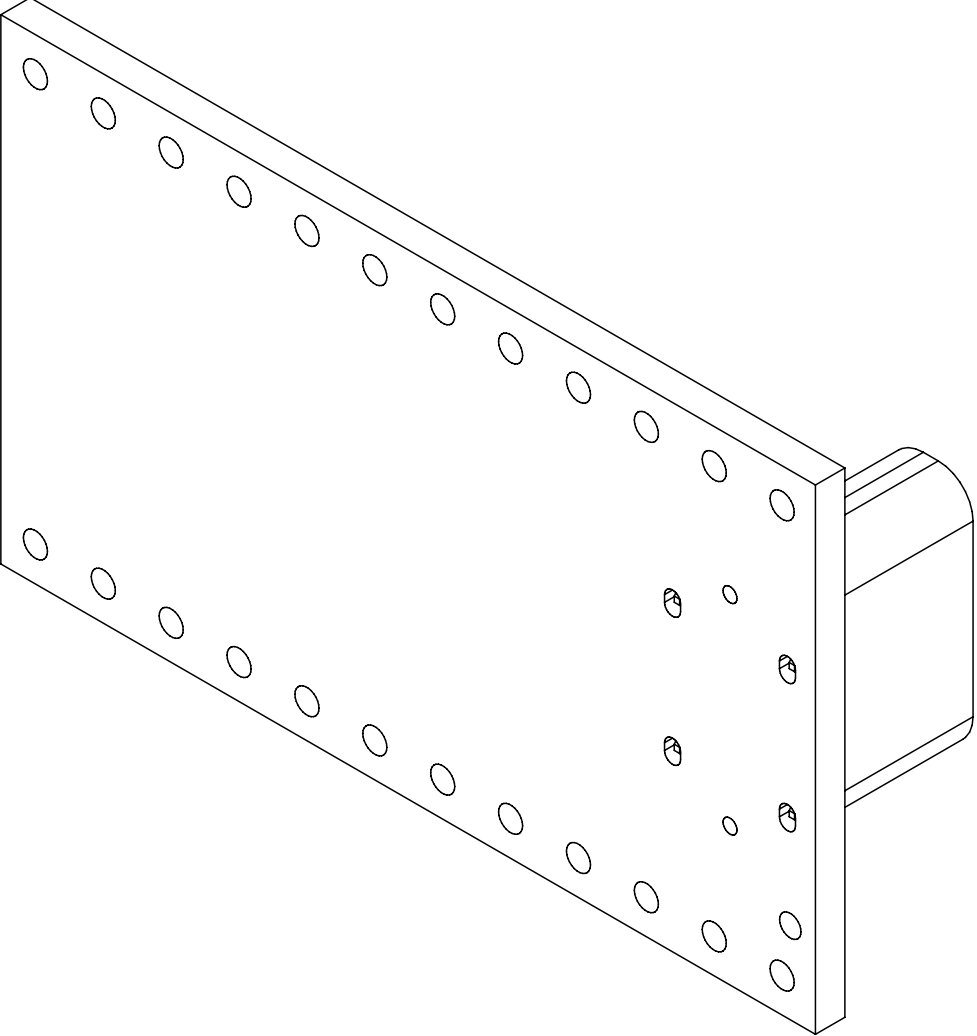
ASSEMBLY DRAWING

3D VIEW

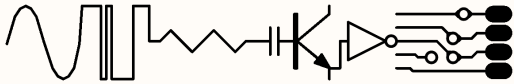
View from Top side (Scale 5:1)



View from Bottom side (Scale 5:1)



SILICONVALLEYGARAGE



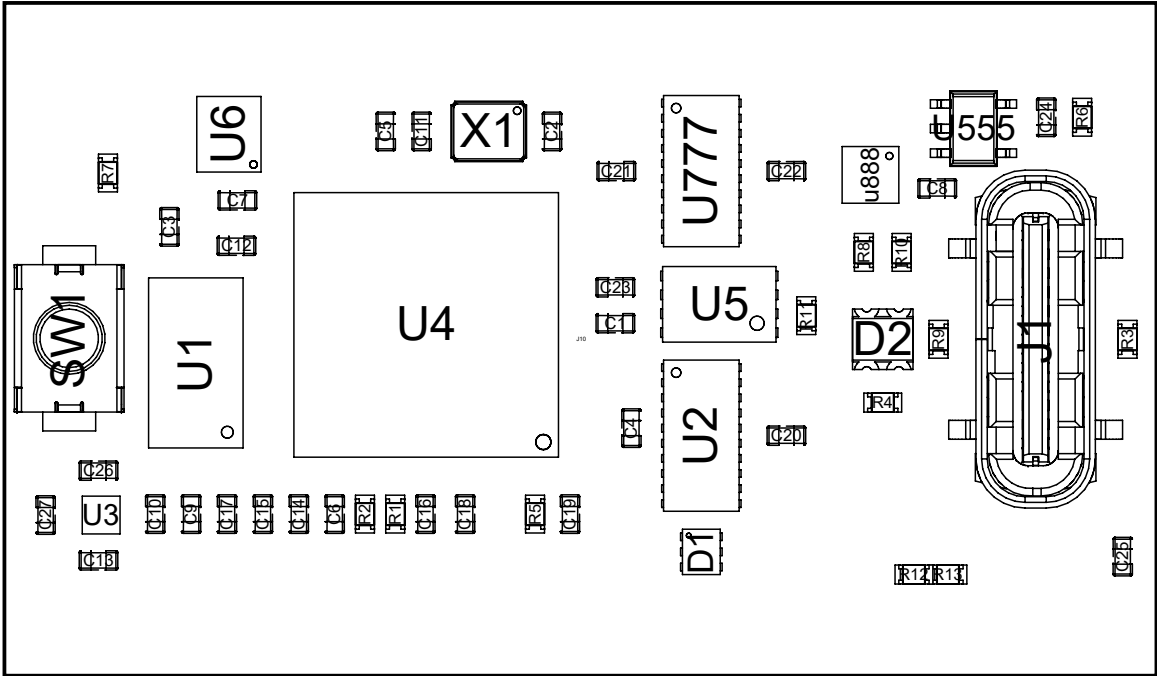
Project PicoRom-2732.PrjPcb

Version: | Variant [No Variations]

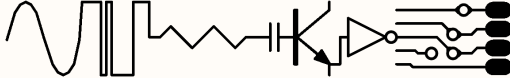
ASSEMBLY DRAWING

1	2	3	4	5	6
Bill Of Materials					
Quantity	Designator	Description	Manufacturer 1	Manufacturer Part Number 1	LCSC
27	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27	CAPACITOR,CERAMIC,100nF,16V,X7R,0402	YAGEO	CC0402KRX7R7BB104	C60474
1	D1				C3700524
1	D2				C601684
1	D3				C552848
1	J1				C2686963
1	J10				
5	R1, R2, R6, R12, R13	RESISTOR,27R,1%,63mW,0402 (1005)			C138021
2	R3, R4	RESISTOR,5K1,1%,63mW,0402 (1005)			C105872
4	R5, R8, R9, R10	RESISTOR,1K,1%,63mW,0402 (1005)			C106235
1	R7	RESISTOR,100K,1%,63mW,0402 (1005)			C60491
1	R11	RESISTOR,10K,1%,63mW,0402 (1005)			C60490
1	SW1	SWITCH,TACT,SMD,2PIN,SMT,TOP ACTUATOR	XUNPU	TS-1088-AR02016	C720477
1	U1	IC,BUS BUFFER,74245, QQFN420X250	NEXPERIA	74AHCT245BQ,115	C554611
2	U2, U777	IC,LEVEL SHIFTER,8CH,DUAL SUPPLY, TI TXB108	TI	TXB0108DQSR	C427057
1	U3				C2860046
1	U4	IC,MICROCONTROLLER,RP2040, PICOROM SYMBOL	RPI	RP2040	C2040
1	U5	IC,FLASH,16Mbit, USON	WINBOND	W25Q16JVUXIQ	C2843335
1	U6	IC,LEVEL SHIFTER,4CH,DUAL SUPPLY, TI TXB104	TI	TXB0104RUTR	C105266
1	U555	IC,BUFFER WITH OE, ONSEMI M74VHC1GT126DF2G	ONSEMI	M74VHC1GT126DF2G	
1	U888	IC,IO EXPANDER, SINGLE WIRE, 5 OUTPUTS, TI TCA5405	TI	TCA5405RUGR	
1	X1	XTAL,4PIN			C669076

DESIGNATORS FRONT



View from Top side (Scale 5:1)

1	2	3	4	5	6
DESIGNATORS BACK					
A					
B					
C					
D	View from Bottom side (Scale 5:1)				
SILICONVALLEYGARAGE 				Project PicoRom-2732.PrjPcb Version: Variant [No Variations] ASSEMBLY DRAWING	
1	2	3	4	5	6